

# React Day 11

---

## What is React Router?

◆ React Router ek library hai jo tumhare React app ko **multi-page banane ka feel** deti hai — bina actual pages reload kare.

## Summary:

| Task          | Code                                                      |
|---------------|-----------------------------------------------------------|
| Install       | <code>npm install react-router-dom</code>                 |
| Define routes | <code>&lt;Routes&gt;&lt;Route /&gt;&lt;/Routes&gt;</code> |
| Add router    | <code>&lt;BrowserRouter&gt;</code> in App                 |
| Navigate      | <code>&lt;Link to="/about"&gt;</code>                     |

[check code for basic understanding of what we are doing](#)

### what is `BrowserRouter` ?

- this enables the client side routing using the browser's history api

### what is `Routes` ?

- it is responsible for rendering the appropriate component based on the current URL

### what is `Route` ?

- each route component defines a path and corresponding component to render when that path is matched.

### what is `Link` ?

- it's for navigate from one page to other page

## Nested routes ke liye outlet kyu ?

- React Router mein `<Outlet />` ek **placeholder** hota hai jahan **nested routes ka component** render hota hai.
  - **Yeh Outlet batata hai ki: "Jis nested route pe tum ho, uska component yahan render karo."**
- 

## 404 Page kaise banaye ?

### ? What is **404**?

◆ **"404"** is an HTTP **status code** that means:

**"The page you are looking for does not exist."**

### 🔧 In React Router (404 Setup):

React Router mein agar koi route **match nahi hota**, to use handle karne ke liye ye likhte hain:

```
<Route path="*" element={<h2>404 - Page Not Found</h2>} />
```

✓ `path="*" means: Match any unknown path`

---

## LayOut Route and Index Route

### 1. What is a **Layout Route**?

A **Layout Route** is a parent route that shares **common UI** (like navbar, footer, sidebar) across its nested child routes.

Real-Life Example:

- `"/dashboard"` → Sidebar + header + main content

- `"/dashboard/profile"` → Same sidebar, new content inside

So instead of repeating navbar/footer on every page, use a layout route with `<Outlet />`

## 2. What is an **Index Route**?

An **Index Route** is the **default child route** that loads when no specific child route is provided.

Socho:

Tum `/dashboard` pe jaate ho — lekin tumne profile, settings kuch bhi specify nahi kiya.

Then automatically, index route load hoti hai.

### Summary Table:

| Feature             | Purpose                                               | Key Tag                       |
|---------------------|-------------------------------------------------------|-------------------------------|
| <b>Layout Route</b> | Shared layout (like navbar/sidebar) for nested routes | <code>&lt;Outlet /&gt;</code> |
| <b>Index Route</b>  | Default nested page (like home inside layout)         | <code>index</code>            |

---

## Route Prefixes

### What Are Route Prefixes?

Route prefixes simply mean:

**The path of nested routes is *relative to their parent route*.**

Matlab:

Nested routes automatically **inherit the parent's path as prefix**.

 Resulting Paths:

| Route Definition | Full Path                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------|
| path="/college"  | /college                                                                                             |
| path="student"   | /college/student  |

---